

## Pyramid Problems · Nested Loops — English Worksheet

**Topic:** Nested Loops · **Course:** Pyramid Problems · **Time:** about 45 minutes

### 1 · Predict 🧠

Read each set of steps. Write what you think will happen.

```
repeat 3 times:
  repeat 4 times:
    place block down
    move forward
  turn right
```

**What shape does the agent leave on the ground? How many blocks?**

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```
repeat 5 times:
  repeat 5 times:
    place block down
    move forward
  turn right
```

**What shape does the agent leave on the ground? How many blocks?**

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```
repeat 4 times:
  repeat 6 times:
    place block down
    move up by 1
  move forward
```

**Is this a flat shape or a 3D shape? Describe what it looks like.**

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## 2 · Spot the Bug 🐛

Each code block below is broken. Rewrite it so it works, then explain the fix.

**Bug A** — The agent is supposed to build a **5 × 5 square** on the ground using a nested loop.

```
repeat 5 times:
  place block down
  move forward
  turn right
```

**Write the fixed code:**

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**Why was it wrong? Why does your fix work?**

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**Bug B** — The agent is supposed to build a **4 × 4 square**. After each side, it should turn right.

```
repeat 4 times:
  repeat 4 times:
    place block down
    move forward
    turn right
```

**Write the fixed code:**

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Why was it wrong? Why does your fix work?

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**Bug C** — The agent is supposed to build a **3-block-thick wall** that is 6 blocks long: 3 rows of 6 blocks, stacked side by side.

```
repeat 6 times:  
  place block down  
  move forward  
move forward
```

Write the fixed code:

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Why was it wrong? Why does your fix work?

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### 3 · Show Your Work 📷 🎥

Switch to your homework world. Build a **7×7 square floor** using a nested loop, then stand on it and check you can see the center.

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

**1 · Your code.** Scroll through it. Say what each part does.

**2 · Your build.** Point the camera. Name the parts.

Fill the blanks:

Today I built \_\_\_\_\_.

I built it using this code: \_\_\_\_\_.

In this code I used \_\_\_\_\_.

The hardest part was \_\_\_\_\_.

That part was hard because \_\_\_\_\_.

The most fun part was \_\_\_\_\_.

Something new I learned was \_\_\_\_\_.

**Write your lines here, then say them in your video.**

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Submit 

Send this worksheet + your video to teacher on KakaoTalk.